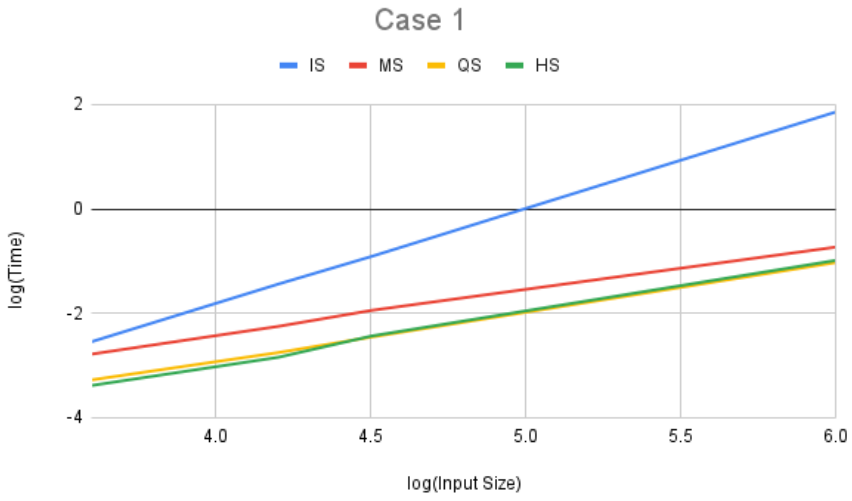
Algorithms PA 1 Report

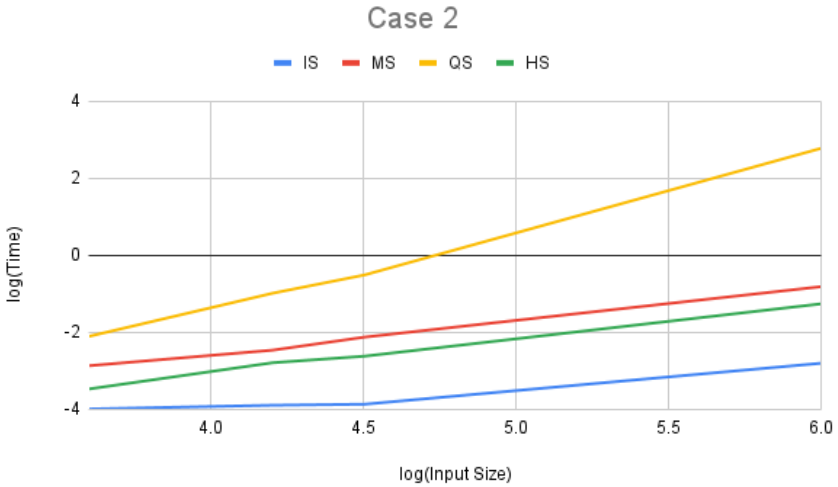
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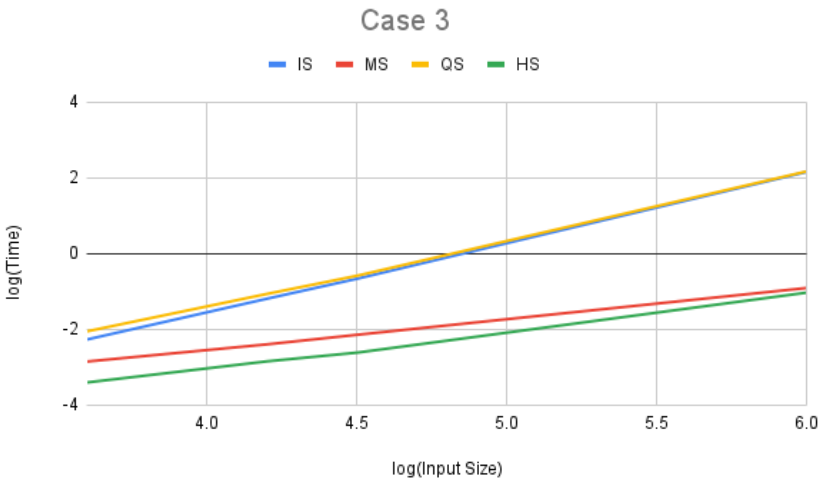
Comparison

Input Size	IS		MS		QS		HS	
	CPU Time (s)	Memory (KB)	CPU Time (s)	Memory (KB)	CPU Time (s)	Memory (KB)	CPU Time (s)	Memory (KB)
4000.case2	0.000103	6068	0.001365	6204	0.007876	6068	0.00034	6068
4000.case3	0.005404	6068	0.001419	6204	0.008882	6068	0.000398	6068
4000.case1	0.002857	6068	0.001654	6204	0.000531	6068	0.000415	6068
16000.case2	0.000129	6220	0.003464	6220	0.104	6320	0.001634	6220
16000.case3	0.0651	6220	0.004052	6220	0.0872	6220	0.001442	6220
16000.case1	0.0363	6220	0.005624	6220	0.001773	6220	0.001438	6220
32000.case2	0.000137	6352	0.007555	6480	0.31	6604	0.002418	6352
32000.case3	0.22	6352	0.007258	6480	0.264	6388	0.002455	6352
32000.case1	0.123	6352	0.0114	6480	0.003519	6352	0.003684	6352
1000000.case2	0.001586	12308	0.154	16400	>600	-	0.05517	12308
1000000.case3	143.22	12308	0.124	16400	148.19	15512	0.09352	12308
1000000.case1	71.38	12308	0.185	16400	0.0938	12308	0.103	12308

Graphs







Discussion

1. In Case 1, the growth of the running time of IS is the largest due to its $\theta(n^2)$ average time complexity, while the 3 other sorts have $\theta(\text{nlgn})$ average time complexity.
2. In Case 2, the growth of the running time of QS is the largest because the rightmost element (as the pivot) of the array is always the largest element. Therefore, it is the worst case for QS and has $\theta(n^2)$ time complexity. Furthermore, IS runs the fastest with an array already sorted. The time complexity is only $\theta(\text{n})$. Finally, the time complexity for MS and HS remains $\theta(\text{nlgn})$.

3. In Case 3, the growth of the running time of QS and IS are both large. For QS, this is still the worst case because the pivot is the smallest element. For IS, this is also considered the worst case, with time complexity $\theta(n^2)$. The time complexity for MS and HS remains $\theta(\mathrm{nlgn})$.